



Measurements of Mixing Enthalpy for a Lead-Free Solder Bi-In-Sn System

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Measurements of mixing enthalpy for a Bi-In-Sn system needs special attention as far as the lead-free solder applications are concerned. Lead-free solder is generally used for solder joints in electronic equipment. In this work, an MHTC 96 Line Evo (Setaram Instruments, France) drop-solution calorimeter was used to determine the integral enthalpy of mixing for a Bi-In-Sn system at 767 K, 813 K and 855 K using drop technique. The enthalpy of mixing was measured for three different cross sections of $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$. Indium was dropped onto the Bi-Sn alloys over the entire composition of indium. Partial quantities of indium mixing were calculated from the integration of the heat flow curve using CALISTO software. Subsequently, integral molar quantities of mixing were calculated from the partial quantities. Mixing enthalpy was found to be almost temperature-independent in the Bi-In-Sn system. Iso-enthalpy curves for the integral mixing enthalpy were plotted at 813 K. The data from this study were analysed with a least-squares fit using the Redlich–Kister–Muggianu polynomial to determine the ternary interaction parameter.

Key words: Drop calorimeter, lead-free solders, Bi-In-Sn system, ternary, enthalpy of mixing, thermodynamic modelling

INTRODUCTION

In modern times, technology is growing very fast. Electronic products are aging within a very small span of time as people adopt new products, and old electronic products are not reused. This results in an accumulation of a huge quantity of electronic waste. The electronic circuits in this equipment are connected by conventional lead-tin solder alloys. The lead in the lead-tin solder alloys are considered worldwide to be very toxic. Mostly, electronic waste is disposed of in landfills where the lead leaches and contaminates groundwater. The presence of lead in drinking water has several health hazards such as increased blood pressure, diminished learning ability in children, and kidney damage. The past few years have seen major changes in soldering

materials for electronics assembly manufacture due to a shift from lead-tin to lead-free soldering. Lead has all the properties required for a solder material, and it is not possible to replace lead by any one metal. Therefore, an alternate lead-free solder is a multi-component system. There is much research into replacing lead with more than one metal. The first step in the search of new lead-free solder alloys is to accurately determine the phase diagram and thermodynamic properties. A thermodynamic database for the lead-free solder systems has been proposed by many researchers working in this area. Experimental thermodynamic data on various lead-free solder systems are very essential to develop a useful thermodynamic database. It would be easier to have thermodynamic data for higher order systems without conducting experiments. Mixing enthalpy is one of the most important properties to accurately optimize phase diagrams. It is quite possible to calculate properties

such as surface tension and viscosity from the thermodynamic data.

It is reported in the literature¹ that Bi-In-Sn-Zn is a potential lead-free solder system. Bi-In-Sn is an important subsystem of this quaternary system. Thermochemical and constitutional data for the four boundary ternary systems of Bi-In-Sn-Zn were optimized by Moelans et al.¹ using calculation of phase diagram (CALPHAD) to obtain a thermodynamic interpretation of the quaternary system. Melting and phase transformation behaviour of the Bi-In-Sn system have been studied by Witusiewicz et al.² using differential scanning calorimetry (DSC) for 29 different alloys. They have found that the phase equilibria data matches quite well with the thermodynamic data. Thermodynamic properties of the Bi-In-Sn system have been measured by Brunetti et al.³ using torsion-effusion (T. E.) and differential scanning calorimetry. They have observed that Bi-In-Sn is a non-ideal system at $x_{\text{Bi}} = 0.20$, and 565 K and the enthalpy of mixing is exothermic at $x_{\text{In}} = 0.58$ and $x_{\text{Sn}} = 0.22$. Activities of components in the Bi-In-Sn ternary system at 1000 K and 1050 K have been predicted by Tao⁴ using a novel molecular interaction volume model (MIVM). They have also calculated the activities of the components in Bi-In-Sn at 550 K and the Bi-In-Sn-Zn quaternary system at 700 K using this model. The chemical reaction between base metal and lead-free solder alloys can be better described by the MIVM to calculate the phase diagram. The MIVM is a very simple statistical thermodynamic model and thermodynamic calculations are based on only two activity coefficients at infinite dilution. Thermodynamic properties of Bi-Sn-Zn,^{5–9} Bi-In-Zn¹⁰ and In-Sn-Zn¹¹ were investigated by calorimetry in the temperature range of 713–773 K. The comparison between the calculated and experimental data are quite satisfactory.

Thermodynamic investigations of In-Zn-Sn have been carried out by Behera and Sonaye¹² using electromotive force (EMF) with molten salt as the electrolyte in the temperature range of 753 K to 853 K for 15 different compositions along three sections. It is reported that the activity of zinc exhibits positive deviation from Raoult's law for all the compositions of zinc. The same technique was employed by Behera and Shamsuddin¹³ for the thermodynamic investigations of the Sn-Zn-Ga system from 723 K to 823 K. Sn-Zn-Ga is a non-ideal system as component activity shows positive deviations. The activity of indium in Bi-In-Sn alloys was measured by Kumar et al.¹⁴ using a galvanic cell assembly fabricated from glass in the temperature range 723 K to 855 K. They used LiCl-KCl and InCl as the electrolyte. In this case, the authors observed that the activity of indium has a positive deviation for most of compositions and negative deviation for few compositions.

The activity of Bi and Sn in the Bi-Sn system measured by various authors^{15–17} indicates good

consistency and shows a very small positive deviation from Raoult's law. Mixing enthalpy of the Bi-Sn system with calorimetric measurement was also reported by a few authors.^{18–20} All the data show an endothermic nature for the enthalpy of mixing. Several lead-free system investigations by different methods are available in the literature.^{21–28} The enthalpy of mixing the liquid In-Bi alloy was measured by Yakymovych et al.²⁹ using calorimetry. Partial molar and integral excess thermodynamic properties of In and Bi in In-Bi liquid alloys for the temperature range of 673–873 K were obtained using EMF (electromotive force) by Gregorczyk et al.³⁰ It is inferred from the literature survey that calorimetric data of the Bi-In-Sn system are very scarce. Therefore, we have carried out calorimetric investigations of the Bi-In-Sn ternary system along three of the cross sections while varying indium content along each of the cross sections, using drop calorimetry at temperatures of 767 K, 813 K and 855 K. Indium was dropped into the Bi-Sn alloys at a regular time intervals. An MHTC 96 Lline Evo (Setaram Instruments, France) calorimeter was used for the enthalpy measurements. Partial and integral thermodynamic properties were calculated from the calorimetric data. Isoenthalpy curves were plotted for the integral enthalpy of mixing. The data from these measurements were treated with the theoretical Redlich-Kister-Muggianu substitutional solution model to derive the ternary interaction parameter by least-squares fitting.

EXPERIMENTAL

Materials

Pure metals (In, Bi, Sn) were used in this study to form binary and ternary alloys. $\alpha\text{-Al}_2\text{O}_3$ needles procured from NIST (National Institute of Standards and Technology, Gaithersburg, MD, USA) were taken as calibration standard. In order to form Bi-Sn alloys, Bi pellets and Sn shots were sealed under vacuum in a quartz tube and heated at 723 K for 48 h with intermediate shaking and, finally, quenching in water. Polishing with fine sand paper made the surface of pure metals free of oxides. Specification details of pure metals and chemicals are provided in Table I.

Calorimetric Measurements

An MHTC 96 Line Evo drop calorimeter (Setaram Instruments, France) was used in this study for binary and ternary systems. It has a thermopile of 20 (Pt-PtRh 10) S-type thermocouples and a continuous water-cooled graphite tube resistance furnace. The maximum operating temperature of the furnace is 1320°C. In order to drop the sample into the crucible automatically, there is a motorized dropping device. The calorimeter was calibrated by dropping $\alpha\text{-Al}_2\text{O}_3$ needles at the end of each

Table I. Materials used in this study

Materials	Source	Initial purity (wt.%)
Indium (ingot)	Johnson Matthey, UK	99.999
Tin (shots)	Johnson Matthey, UK	99.999
Bismuth(lumps)	Johnson Matthey, UK	99.999
Argon gas	Indian oxygen limited, India	> 99

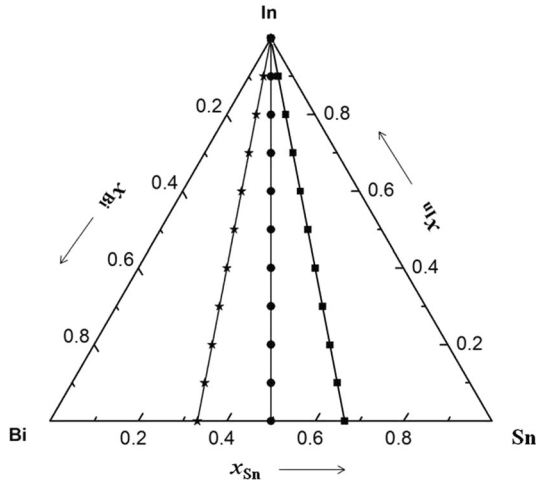


Fig. 1. Compositions of the investigated cross sections along which mixing enthalpy were measured at three cross-sections: $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$; asterisk, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$; circle, $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$; square.

experiment. These needles meet NIST (National Institute of Standards and Technology, Gaithersburg, MD, USA) standard specifications. The temperature of the sample dispenser was controlled by constantan-copper thermocouple before each drop. The calorimeter is totally controlled by CALISTO software. Before the start of the experiments, a certain quantity of binary alloy was taken in the calorimeter crucible (length 60 mm, outer diameter 12 mm). It was used as the solvent for the metal to be dropped from the sample dispenser. The water cooling system and flow of gas was controlled by the CALISTO software. The heating rate, holding time, cooling rate of the furnace and gas flow rate were programmed before the start of the experiments. At the beginning of experiment, the protection tube was evacuated and flushed with high-purity argon (99.9999 vol.%). Oxidation of the alloy was prevented by the continuous flow of argon (approx. 30 mL/min) during the experiments. Weight of the crucible with the sample before and after the experiment was measured to assess weight loss, which was found to be negligible. The recorded heat signals returned to baseline after 30 min of indium drop and 25 min after $\alpha\text{-Al}_2\text{O}_3$ addition. Calorimetric measurements were carried

out for the Bi-In-Sn system along three sections of $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$, shown in Fig. 1, at 767 K, 813 K and 855 K. All experiments were conducted twice keeping all parameters constant. Recorded heat signals were then integrated to evaluate the enthalpy values using CALISTO software (Setaram Instruments, France).

RESULTS AND DISCUSSION

The peaks of the heat flow curve were integrated using CALOSTO data processing software. The calibration constant (K) was evaluated from the integration of the heat flow peaks corresponding to the $\alpha\text{-Al}_2\text{O}_3$ needle drops. The integrated values of the heat signal were then converted to enthalpy after multiplying by the calibration constant. The enthalpy of reaction ($\Delta H_{\text{Reaction,In},i}$) was obtained from the heat signal ($\Delta H_{\text{Signal,In},i}$) and enthalpy increments of indium ($\Delta H_{\text{In},i}^{T_D-T_M}$) from drop temperature (T_D) to bath temperature (T_M) using Eq. 1.

$$\Delta H_{\text{Reaction,In},i} = (\Delta H_{\text{Signal,In},i} \cdot K) - \left(\Delta H_{\text{In},i}^{T_D-T_M} \cdot n_{\text{In},i} \right) \quad (1)$$

where $n_{\text{In},i}$ are the number of moles of indium dropped in the solvent from the top of the calorimeter. $\Delta H_{\text{Signal,In},i}$ is the heat signal from one drop of indium to the solvent. The calculation of the increase in molar enthalpy of indium was obtained from the heat capacity and latent heat of transformation in the above temperature range. The heat capacity data of indium was taken from Kubaschewski and Alcock.³¹

Since a small amount of indium was dropped into the liquid bath, the partial enthalpy ($\Delta \bar{H}_{\text{In},i}^m$) can be directly calculated as:

$$\Delta \bar{H}_{\text{In},i} \approx \Delta H_{\text{Reaction,In},i} / n_{\text{In},i} \quad (2)$$

In each measurement, certain quantities of the liquid Bi-Sn binary alloys were used. Therefore, the enthalpy of mixing binary alloys should be considered for the calculation of the mixing enthalpy. The integral molar enthalpy of mixing ($\Delta_{\text{mix}}H$) was calculated by (Eq. 3) as given below:

$$\Delta_{\text{mix}}H = \left[\frac{\{\Delta_{\text{mix}}H_{\text{Bi-Sn}} \times (n_{\text{Bi}} + n_{\text{Sn}})\} + \sum_i \Delta H_{\text{Reaction,In},i}}{(n_{\text{In}} + n_{\text{Bi}} + n_{\text{Sn}})} \right] \quad (3)$$

The enthalpy of mixing values for the Bi-Sn binary system ($\Delta_{\text{mix}}H_{\text{Bi-Sn}}$) were taken from our experimental work given in Table II by corresponding interpolation. Integral molar enthalpies calculated by Eq. 3 for the Bi-In-Sn ternary system for different compositions along three sections are given in Table III. Partial mixing enthalpies for

indium and mole fraction of indium dropped are also given in Table III.

There are various sources of errors in the calorimetric measurements: (1) calorimeter type based on construction, (2) calibration, (3) integration of heat flow data, (4) degree of dissolution of solute in the solvent, (5) impurity contents. The overall experimental uncertainty of the calorimeter is 10–12%. The error involved in calibration by dropping α - Al_2O_3 needles was estimated to be less than $\pm 1.5\%$.

Enthalpies of mixing were calculated at 767 K, 813 K and 855 K for the given compositions along three sections as shown in Figs. 2, 3 and 4. It is observed that measured enthalpies of mixing in

Table II. Integral molar enthalpy of mixing for the Bi-Sn system; Tin was dropped

No. of moles of tin (mole)	mole fraction (n_{Sn}) x_{Sn}	Standard uncertainties u (x_{Sn})	Heat effect $\Delta H_{\text{Signal}} \cdot K$ (J)	Standard uncertainties $\Delta H_{\text{Signal}} \cdot K$ (J)	Heat of reaction $\Delta H_{\text{Reaction}}$ (J)	Integral enthalpy $\Delta_{\text{mix}}H(\text{Jmole}^{-1})$
Series 1: Bi-Sn alloys; atmosphere: argon at pressure $p = 0.1$ MPa; starting amount $n_{\text{Bi}} = 0.002153$ mol, $K = 0.0036571 \text{ J } \mu\text{V s}^{-1}$, $T_{\text{D}} = 298 \text{ K}$, $T_{\text{M}} = 767 \text{ K}$, $H_{\text{Sn}}^{T_{\text{D}} \rightarrow T_{\text{M}}} = 20581.60 \text{ J mol}^{-1}$ ¹³¹						
0.000232	0.0973	0.0001	4.836	0.007	0.061	25.58
0.000310	0.2011	0.0001	6.468	0.025	0.088	55.29
0.000418	0.3084	0.0001	8.721	0.016	0.118	85.77
0.000530	0.4090	0.0002	11.04	0.018	0.130	109.0
0.000689	0.5030	0.0002	14.32	0.067	0.140	124.0
0.001102	0.6038	0.0001	22.85	0.066	0.170	130.1
0.001691	0.6978	0.0001	34.94	0.154	0.140	118.88
0.003561	0.7985	0.0002	73.4	0.133	0.110	89.56
0.005028	0.8630	0.0002	103.53	0.181	0.030	62.81
Series 2: Bi-Sn alloys; atmosphere: argon at pressure $p = 0.1$ MPa; starting amount $n_{\text{Bi}} = 0.002153$ mol, $K = 0.0037225 \text{ J } \mu\text{V s}^{-1}$, $T_{\text{D}} = 298 \text{ K}$, $T_{\text{M}} = 813 \text{ K}$, $H_{\text{Sn}}^{T_{\text{D}} \rightarrow T_{\text{M}}} = 21843.09 \text{ J mol}^{-1}$ ¹³¹						
0.000245	0.1022	0.0001	5.416	0.022	0.064	26.69
0.000314	0.2061	0.0001	6.953	0.030	0.094	58.26
0.000364	0.3001	0.0002	8.06	0.016	0.109	86.80
0.000537	0.4041	0.0002	11.87	0.024	0.140	112.6
0.000705	0.5014	0.0001	15.55	0.081	0.150	129.0
0.001138	0.6054	0.0001	25.03	0.099	0.170	133.2
0.001704	0.6993	0.0002	37.38	0.169	0.160	123.88
0.003785	0.8033	0.0002	82.8	0.153	0.120	92.01
0.004853	0.8637	0.0001	106.03	0.178	0.030	65.64
Series 3: Bi-Sn alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; starting amount $n_{\text{Bi}} = 0.002153$ mol, $K = 0.00380285 \text{ J } \mu\text{V s}^{-1}$, $T_{\text{D}} = 298 \text{ K}$, $T_{\text{M}} = 855 \text{ K}$, $H_{\text{Sn}}^{T_{\text{D}} \rightarrow T_{\text{M}}} = 22976.33 \text{ J mol}^{-1}$ ¹³¹						
0.000220	0.0927	0.0001	5.122	0.011	0.067	28.23
0.000274	0.1866	0.0001	6.38	0.038	0.081	55.91
0.000388	0.2906	0.0001	9.029	0.024	0.114	86.3
0.000463	0.3845	0.0001	10.760	0.027	0.120	109.2
0.001007	0.5221	0.0002	23.35	0.103	0.210	131.4
0.000958	0.6059	0.0002	22.16	0.042	0.150	135.8
0.001403	0.6864	0.0002	32.37	0.141	0.130	127.0
0.002238	0.7635	0.0002	51.55	0.230	0.130	110.06
0.002258	0.8105	0.0001	51.93	0.098	0.050	92.59
0.004275	0.8623	0.0002	98.25	0.1571	0.030	69.19

^aStandard uncertainties u are: $u(T_{\text{M}}) = 1 \text{ K}$; $u(p) = 10 \text{ kPa}$; $u(n_{\text{Bi}}) = 0.000001 \text{ mol}$, $u(n_{\text{Sn}}) = 0.000002 \text{ mol}$; $u(\Delta H_{\text{Reaction}}) = 0.001 \text{ kJ}$; Series 1: $u(K) = 0.00001 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}}H) = 0.0018 \text{ kJ mol}^{-1}$; Series 2: $u(K) = 0.00002 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}}H) = 0.0023 \text{ kJ mol}^{-1}$; Series 3: $u(K) = 0.00001 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}}H) = 0.0026 \text{ kJ mol}^{-1}$.

^b T_{D} = temperature at which tin was dropped in Kelvin, T_{M} = temperature of the liquid alloy bath in Kelvin, n_{Sn} = no of moles of tin, K = calibration constant, ΔH_{Signal} = The heat signal due to drop of tin to the bath.

Table III. Partial and integral mixing enthalpies of Bi-In-Sn alloys

No. of moles of indium n_{In} (mol)	Mole fraction (x_{In})	Standard uncertainties $u(x_{\text{In}})$	Heat effect $\Delta H_{\text{Signal}} \cdot K$ (J)	Standard uncertainties $\Delta H_{\text{Signal}} \cdot K$ (J)	Heat of Reaction $\Delta H_{\text{Reaction}}$ (J)	Partial enthalpy ΔH_{In} (J mol ⁻¹)	Integral enthalpy $\Delta_{\text{mix}}H$ (J mol ⁻¹)
Series 1: (Sn _{0.33} Bi _{0.67}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.002088$ mol, $n_{\text{Sn}} = 0.00104$ mol, $K = 0.00367$ J $\mu\text{V s}^{-1}$, $T_{\text{D}} = 298$ K, $T_{\text{M}} = 767$ K, $\Delta_{\text{mix}}H_{\text{Bi-Sn}} = 90.13$ J mol ⁻¹ , $H_{\text{In}}^{T_{\text{D}} \rightarrow T_{\text{M}}} = 16906.14$ (J mol ⁻¹)							
0.000340	0.0980	0.0001	4.147	0.01	-1.601	-4709	-380.4
0.000508	0.2133	0.0001	6.561	0.01	-2.027	-3990	-841.6
0.000545	0.3081	0.0001	7.398	0.01	-1.815	-3330	-1142
0.000783	0.4103	0.0001	11.26	0.02	-1.976	-2524	-1346
0.000880	0.4942	0.0001	13.27	0.02	-1.602	-1820	-1413
0.001687	0.6026	0.0002	26.33	0.05	-2.19	-1298	-1389
0.002927	0.7103	0.0001	47.16	0.10	-2.32	-792.6	-1227
0.005452	0.8075	0.0002	89.83	0.20	-2.334	-428.1	-959.0
0.005351	0.8552	0.0001	88.93	0.20	-1.535	-286.9	-792.5
0.005453	0.8844	0.0001	91.55	0.20	-0.637	-116.8	-656.3
0.005417	0.9037	0.0001	90.97	0.20	-0.607	-112.1	-565.5
Series 2: (Sn _{0.33} Bi _{0.67}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.002088$ mol, $n_{\text{Sn}} = 0.00104$ mol, $K = 0.00327$ J $\mu\text{V s}^{-1}$, $T_{\text{D}} = 298$ K, $T_{\text{M}} = 813$ K, $\Delta_{\text{mix}}H_{\text{Bi-Sn}} = 94.15$ J mol ⁻¹ , $H_{\text{In}}^{T_{\text{D}} \rightarrow T_{\text{M}}} = 18249.02$ (J mol ⁻¹)							
0.000361	0.1035	0.0002	5.095	0.01	-1.493	-4136	-343.5
0.00047	0.2099	0.0001	6.773	0.01	-1.804	-3838	-758.4
0.000497	0.2980	0.0001	7.394	0.01	-1.676	-3372	-1050
0.000932	0.4195	0.0001	14.66	0.03	-2.348	-2519	-1304
0.000875	0.5006	0.0001	14.44	0.03	-1.528	-1746	-1366
0.001993	0.6211	0.0002	34.14	0.08	-2.232	-1120	-1307
0.002349	0.7050	0.0001	41.21	0.09	-1.658	-705.8	-1173
0.005132	0.8012	0.0002	91.52	0.21	-2.134	-415.8	-926.4
0.00532	0.8515	0.0001	96.13	0.22	-0.956	-179.7	-737.7
0.005463	0.8821	0.0001	99.14	0.24	-0.553	-101.2	-606.6
0.005447	0.9021	0.0001	98.93	0.23	-0.470	-86.29	-518.0
Series 3: (Sn _{0.33} Bi _{0.67}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.002088$ mol, $n_{\text{Sn}} = 0.00104$ mol, $K = 0.00333$ J $\mu\text{V s}^{-1}$, $T_{\text{D}} = 298$ K, $T_{\text{M}} = 855$ K, $\Delta_{\text{mix}}H_{\text{Bi-Sn}} = 96.44$ J mol ⁻¹ , $H_{\text{In}}^{T_{\text{D}} \rightarrow T_{\text{M}}} = 19473.15$ (J mol ⁻¹)							
0.000347	0.0999	0.0001	5.430	0.01	-1.327	-3824	-295.1
0.000432	0.1994	0.0002	6.863	0.01	-1.549	-3586	-658.9
0.000629	0.3104	0.0001	10.22	0.02	-2.025	-3219	-1014
0.000871	0.4215	0.0001	14.96	0.03	-2.005	-2302	-1221
0.000983	0.5105	0.0002	17.47	0.04	-1.671	-1700	-1295
0.001596	0.6083	0.0001	29.35	0.08	-1.733	-1086	-1253
0.002811	0.7103	0.0001	52.94	0.13	-1.802	-641.1	-1094
0.004488	0.7954	0.0002	85.87	0.22	-1.517	-338.0	-871.9
0.004655	0.8431	0.0002	89.68	0.21	-0.958	-205.8	-716.4
0.003976	0.8692	0.0001	77.28	0.21	-0.151	-37.98	-603.6
0.004263	0.8890	0.0001	82.66	0.20	-0.354	-83.04	-524.9
0.004163	0.9033	0.0001	81.32	0.22	0.251	60.39	-449.5

Table III. continued

No. of moles of indium n_{In} (mol)	Mole fraction (x_{In})	Standard uncertainties u (x_{In})	Heat effect $\Delta H_{\text{Signal}} \cdot K$ (J)	Standard uncertainties $\Delta H_{\text{Signal}} \cdot K$ (J)	Heat of Reaction $\Delta H_{\text{Reaction}}$ (J)	Partial enthalpy $\Delta \bar{H}_{\text{In}}$ (J mol ⁻¹)	Integral enthalpy $\Delta_{\text{mix}} H$ (J mol ⁻¹)
Series 4: (Sn _{0.50} Bi _{0.50}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.001699$ mol, $n_{\text{Sn}} = 0.00374$ J $\mu\text{V s}^{-1}$, $T_D = 298$ K, $T_M = 767$ K, $\Delta_{\text{mix}} H_{\text{Bi-Sn}} = 122.26$ J mol ⁻¹ , $H_{\text{In}}^{T_D-T_M} = 16906.20$ (J mol ⁻¹)							
0.000405	0.1065	0.0001	5.481	0.01	-1.366	-3373	-250.0
0.000443	0.1997	0.0001	6.359	0.02	-1.13	-2551	-490.0
0.000619	0.3015	0.0002	9.017	0.02	-1.448	-2339	-725.3
0.001005	0.4211	0.0002	14.78	0.04	-2.213	-2202	-978.1
0.001108	0.5130	0.0002	17.06	0.04	-1.675	-1512	-1063
0.001712	0.6090	0.0002	26.88	0.07	-2.066	-1207	-1091
0.00261	0.6993	0.0001	42.49	0.11	-1.637	-627.2	-984.0
0.005447	0.7971	0.0002	90.11	0.23	-1.975	-362.6	-781.9
0.005346	0.8462	0.0001	89.37	0.23	-1.011	-189.1	-638.5
0.00542	0.8765	0.0001	91.38	0.23	-0.253	-46.68	-521.9
0.004909	0.8952	0.0002	82.85	0.22	-0.14	-28.52	-447.2
0.004153	0.9071	0.0001	70.13	0.18	-0.083	-19.99	-398.7
Series 5: (Sn _{0.50} Bi _{0.50}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.001699$ mol, $n_{\text{Sn}} = 0.001699$ mol, $K = 0.00332$ J $\mu\text{V s}^{-1}$, $T_D = 298$ K, $T_M = 813$ K, $\Delta_{\text{mix}} H_{\text{Bi-Sn}} = 126.85$ J mol ⁻¹ , $H_{\text{In}}^{T_D-T_M} = 18249.02$ (J mol ⁻¹)							
0.000358	0.0953	0.0001	5.412	0.01	-1.121	-3131	-183.7
0.000525	0.2063	0.0001	8.324	0.02	-1.257	-2394	-454.8
0.000654	0.3114	0.0001	10.38	0.02	-1.554	-2376	-709.4
0.000764	0.4038	0.0001	12.28	0.03	-1.664	-2178	-906.3
0.001073	0.4982	0.0001	17.86	0.04	-1.72	-1603	-1017
0.001817	0.6044	0.0001	30.89	0.07	-2.268	-1248	-1066
0.00326	0.7132	0.0001	57.84	0.14	-1.652	-506.7	-911.9
0.005542	0.8046	0.0002	99.26	0.24	-1.88	-339.2	-729.4
0.005492	0.8515	0.0002	99.97	0.24	-0.251	-45.70	-565.3
0.005988	0.8823	0.0002	109	0.24	-0.271	-45.26	-457.4
0.005076	0.8999	0.0001	92.82	0.25	0.192	37.83	-383.4
Series 6: (Sn _{0.50} Bi _{0.50}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.001699$ mol, $n_{\text{Sn}} = 0.001699$ mol, $K = 0.00332$ J $\mu\text{V s}^{-1}$, $T_D = 298$ K, $T_M = 855$ K, $\Delta_{\text{mix}} H_{\text{Bi-Sn}} = 127.43$ J mol ⁻¹ , $H_{\text{In}}^{T_D-T_M} = 19473.15$ (J mol ⁻¹)							
0.000383	0.1013	0.0001	6.389	0.02	-1.069	-2791	-168.2
0.000476	0.2018	0.0001	8.16	0.02	-1.109	-2330	-409.9
0.000589	0.2988	0.0002	10.15	0.03	-1.319	-2239	-632.3
0.000847	0.4031	0.0002	14.59	0.04	-1.9	-2243	-871.9
0.001406	0.5213	0.0002	25.23	0.07	-2.145	-1526	-1001.4
0.001686	0.6132	0.0001	31.08	0.09	-1.755	-1041	-1009
0.005285	0.7585	0.0001	100.6	0.28	-2.332	-441.2	-795.7
0.005503	0.8264	0.0001	106.7	0.3	-0.473	-85.95	-596.2
0.005448	0.8642	0.0001	105.8	0.3	-0.34	-62.41	-480.0
0.00559	0.8890	0.0002	108.5	0.31	-0.303	-54.20	-402.2
0.005575	0.9061	0.0001	108.3	0.3	-0.229	-41.08	-346.6

Table III. continued

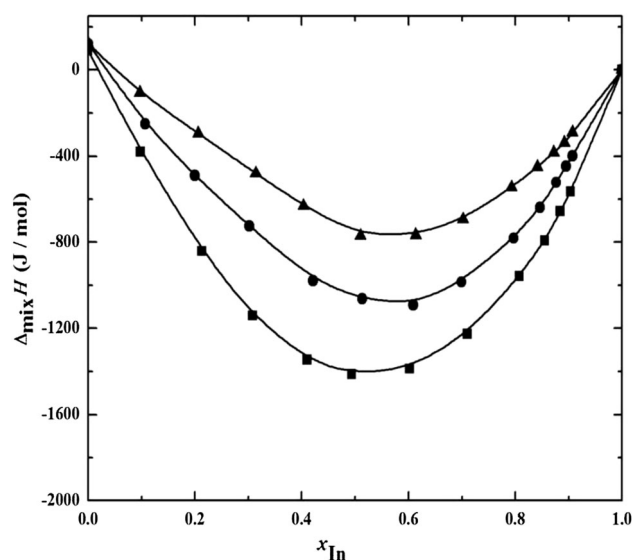
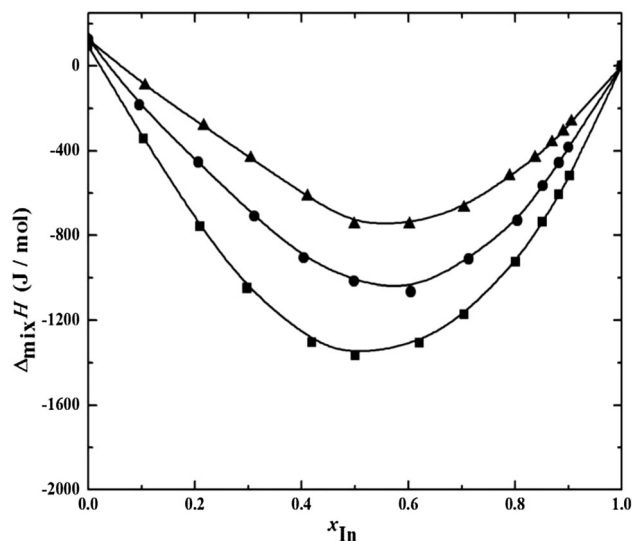
No. of moles of in- dium n_{In} (mol)	Mole fraction (x_{In})	Standard uncertainties u (x_{In})	Heat effect $\Delta H_{\text{Signal}} \cdot K$ (J)	Standard uncertainties $\Delta H_{\text{Signal}} \cdot K$ (J)	Heat of Reaction $\Delta H_{\text{Reaction}}$ (J)	Partial enthalpy $\Delta \bar{H}_{\text{In}}$ (J mol ⁻¹)	Integral enthalpy $\Delta_{\text{mix}} H$ (J mol ⁻¹)
Series 7: (Sh _{0.67} Bi _{0.33}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.001238$ mol, $n_{\text{Sn}} = 0.002475$ mol, $K = 0.00370$ J $\mu\text{V s}^{-1}$, $T_{\text{D}} = 298$ K, $T_{\text{M}} = 767$ K, $\Delta_{\text{mix}} H_{\text{Bi-Sn}} = 121.69$ J mol ⁻¹ , $H_{\text{In}}^{T_{\text{D}}-T_{\text{M}}} = 16906.14$ (J mol ⁻¹)							
0.000396	0.0964	0.0001	5.811	0.02	-0.884	-2232	-105.2
0.000568	0.2061	0.0001	8.656	0.03	-0.947	-1667	-294.9
0.000737	0.3142	0.0001	11.24	0.03	-1.219	-1654	-479.9
0.000807	0.4032	0.0001	12.31	0.04	-1.331	-1649	-631.6
0.001354	0.5098	0.0001	20.98	0.06	-1.911	-1411	-771.0
0.002033	0.6136	0.0001	32.84	0.09	-1.533	-754.1	-767.4
0.002845	0.7018	0.0002	46.83	0.14	-1.272	-447.1	-694.2
0.005496	0.7931	0.0002	91.77	0.28	-1.143	-208.0	-545.3
0.005466	0.8414	0.0002	91.65	0.27	-0.761	-139.2	-450.5
0.00562	0.8721	0.0001	94.42	0.29	-0.592	-105.3	-383.7
0.005414	0.8922	0.0001	91.01	0.27	-0.516	-95.31	-338.4
0.005612	0.9073	0.0001	94.9	0.28	0.024	4.277	-290.4
Series 8: (Sh _{0.67} Bi _{0.33}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.001238$ mol, $n_{\text{Sn}} = 0.002475$ mol, $K = 0.00328$ J $\mu\text{V s}^{-1}$, $T_{\text{D}} = 298$ K, $T_{\text{M}} = 813$ K, $\Delta_{\text{mix}} H_{\text{Bi-Sn}} = 125.71$ J mol ⁻¹ , $H_{\text{In}}^{T_{\text{D}}-T_{\text{M}}} = 18249.02$ (J mol ⁻¹)							
0.000441	0.1062	0.0001	7.196	0.02	-0.852	-1932	-92.74
0.000584	0.2163	0.0001	9.7	0.03	-0.957	-1639	-283.3
0.000598	0.3042	0.0001	9.937	0.03	-0.976	-1632	-434.5
0.00096	0.4103	0.0002	15.96	0.05	-1.559	-1624	-615.8
0.001109	0.4986	0.0001	18.58	0.06	-1.66	-1497	-747.8
0.001907	0.6013	0.0002	33.37	0.11	-1.431	-750.4	-748.3
0.003238	0.7041	0.0002	57.66	0.20	-1.429	-441.3	-669.1
0.0051	0.7896	0.0002	92.27	0.30	-0.8	-156.9	-521.1
0.005161	0.8372	0.0001	93.5	0.31	-0.683	-132.3	-433.1
0.005559	0.8691	0.0001	101.1	0.33	-0.346	-62.24	-360.5
0.005452	0.8902	0.0001	99.27	0.33	-0.223	-40.90	-308.9
0.005393	0.9053	0.0001	98.54	0.32	0.123	22.81	-263.3
Series 9: (Sh _{0.67} Bi _{0.33}) _{1-x} In _x alloys; Atmosphere: Argon at pressure $p = 0.1$ MPa; Starting amount $n_{\text{Bi}} = 0.001238$ mol, $n_{\text{Sn}} = 0.002475$ mol, $K = 0.00318$ J $\mu\text{V s}^{-1}$, $T_{\text{D}} = 298$ K, $T_{\text{M}} = 855$ K, $\Delta_{\text{mix}} H_{\text{Bi-Sn}} = 128.00$ J mol ⁻¹ , $H_{\text{In}}^{T_{\text{D}}-T_{\text{M}}} = 19473.15$ (J mol ⁻¹)							
0.000434	0.1047	0.0001	7.709	0.02	-0.742	-1710	-64.32
0.000568	0.2125	0.0001	10.14	0.03	-0.919	-1618	-251.5
0.000592	0.3004	0.0001	10.54	0.03	-0.988	-1669	-409.6
0.000843	0.3963	0.0001	15.05	0.04	-1.363	-1617	-575.1
0.001434	0.5104	0.0001	25.9	0.07	-2.028	-1414	-733.7
0.001835	0.6058	0.0002	34.41	0.10	-1.327	-723.2	-731.7
0.003217	0.7062	0.0001	61.4	0.18	-1.248	-387.9	-644.2
0.005497	0.7952	0.0001	106.4	0.31	-0.661	-120.2	-485.3
0.005491	0.8428	0.0001	106.4	0.30	-0.541	-98.52	-395.4
0.005187	0.8711	0.0001	100.9	0.30	-0.124	-23.91	-328.5

Table III. continued

No. of moles of indium n_{In} (mol)	Mole fraction (x_{In})	Standard uncertainties u (x_{In})	Heat effect $\Delta H_{\text{Signal}} \cdot K$ (J)	Standard uncertainties $\Delta H_{\text{Signal}} \cdot K$ (J)	Heat of Reaction $\Delta H_{\text{Reaction}}$ (J)	Partial enthalpy $\Delta \bar{H}_{\text{In}}$ (J mol ⁻¹)	Integral enthalpy $\Delta_{\text{mix}} H$ (J mol ⁻¹)
0.005703	0.8924	0.0002	111.3	0.32	0.241	42.26	-267.3
0.005548	0.9073	0.0001	108.5	0.31	0.493	88.86	-218.0

^aStandard uncertainties u are: $u(T_{\text{M}}) = 1 \text{ K}$, $u(p) = 10 \text{ kPa}$, $u(n_{\text{Bi}}) = 0.000001 \text{ mol}$, $u(n_{\text{Sn}}) = 0.000002 \text{ mol}$, $u(\Delta H_{\text{Reaction}}) = 0.001 \text{ kJ}$; Series 1: $u(K) = 0.00002 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0233 \text{ kJ mol}^{-1}$; Series 2: $u(K) = 0.00002 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0197 \text{ kJ mol}^{-1}$; Series 3: $u(K) = 0.00002 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.01608 \text{ kJ mol}^{-1}$; Series 4: $u(K) = 0.00001 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.01817 \text{ kJ mol}^{-1}$; Series 5: $u(K) = 0.00001 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0144 \text{ kJ mol}^{-1}$; Series 6: $u(K) = 0.00001 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0127 \text{ kJ mol}^{-1}$; Series 7: $u(K) = 0.00001 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0132 \text{ kJ mol}^{-1}$; Series 8: $u(K) = 0.00002 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0105 \text{ kJ mol}^{-1}$; Series 9: $u(K) = 0.00002 \text{ J } \mu\text{V s}^{-1}$, $u(\Delta_{\text{mix}} H) = 0.0079 \text{ kJ mol}^{-1}$.

^b T_{D} = Drop temperature in Kelvin, T_{M} = Bath temperature in Kelvin, n_{Bi} = number of moles of bismuth, K = Calibration constant, $\Delta_{\text{mix}} H_{\text{Bi-Sn}}$ = Integral enthalpy of the binary Bi-Sn system, ΔH_{Signal} = the heat effect due to single drop of indium sample to the bath, $\Delta_{\text{mix}} H$ = The integral enthalpy of mixing of Bi-In-Sn system.

Fig. 2. Mixing enthalpies of liquid Bi-In-Sn alloys along the cross section of $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$; square, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$; circle and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$; triangle at 767 K.Fig. 3. Mixing enthalpies of liquid Bi-In-Sn alloys along the cross section of $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$; square, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$; circle and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$; triangle at 813 K.

these three cross sections are exothermic in nature. So, the segregation of atoms in this system is ruled out.

It is observed that the nature of enthalpies of mixing variation with indium composition is similar in Figs. 2, 3 and 4. The minima for the enthalpy of mixing for the cross sections $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$ is close to $x_{\text{In}} = 0.55$ and for the cross section $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$, it is almost symmetric. As indium content increases, the enthalpy of mixing decreases, approaching a minimum at $x_{\text{In}} = 0.55$ and again increases towards zero. It can also be inferred that there is no chance of compound

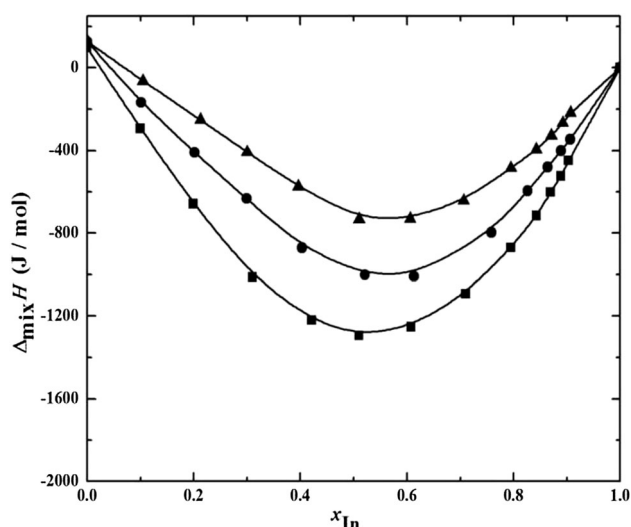


Fig. 4. Mixing enthalpies of liquid Bi-In-Sn alloys along the cross section of $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$; square, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$; circle and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$; triangle at 855 K.

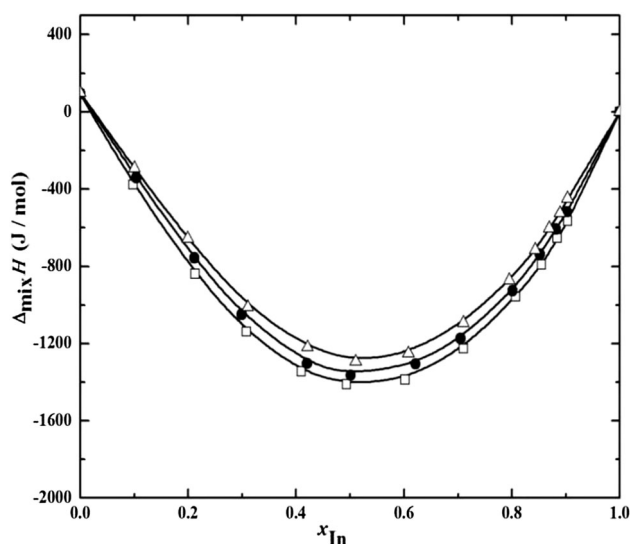


Fig. 5. Mixing enthalpies of liquid Bi-In-Sn alloys along the cross section of $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$ at 767 K; square, 813 K; circle and 855 K; triangle, bold line trend line.

formation in this system due to the absence of a strong interaction among the atoms in this ternary system. It is also seen from Figs. 2, 3 and 4 that the minima in enthalpy of mixing curve become less exothermic as the $x_{\text{Sn}}/x_{\text{Bi}}$ ratio increases. It is a maximum at equiatomic percent and diminishes at the terminal ends. This may be due to a decrease in interatomic attraction among the In, Sn and Bi atoms. The enthalpy of mixing contributes negatively to the Gibbs energy irrespective of the sign of the entropy of mixing. The observed negative enthalpy of mixing infers that finite interatomic interaction energy is present among the atoms in

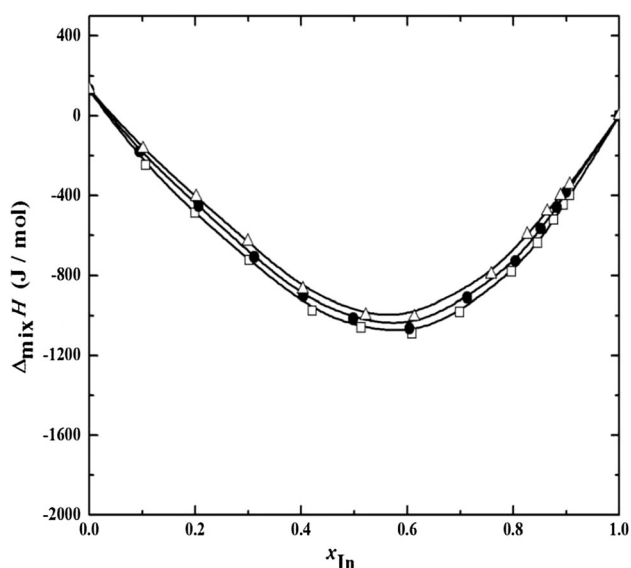


Fig. 6. Mixing enthalpies of liquid Bi-In-Sn alloys along the cross section of $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ at 767 K; square, 813 K; circle and 855 K; triangle, bold line trend line.

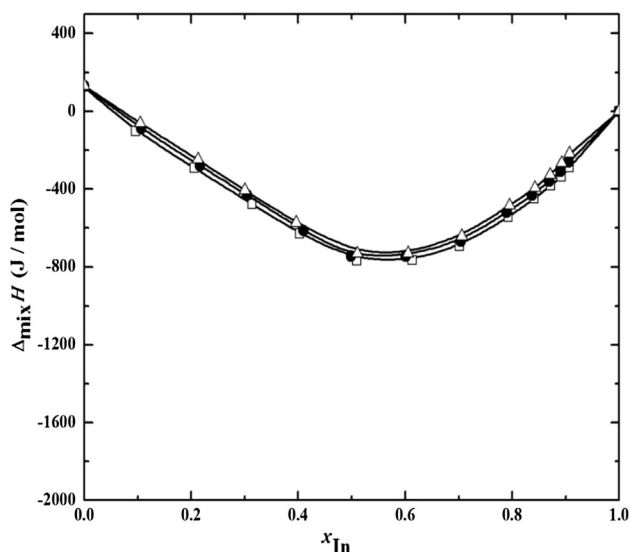


Fig. 7. Mixing enthalpies of liquid Bi-In-Sn alloys along the cross section of $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$ at 767 K; square, 813 K; circle and 855 K; triangle, bold line trend line.

this system. This energy is released in the process of mixing, as confirmed by the exothermic nature of the heat of mixing. From Figs. 5, 6 and 7, it can also be observed that the plots are very close to each other at 767 K, 813 K and 855 K. This may be interpreted that the enthalpy of mixing in the Bi-In-Sn system is almost independent of temperature.

The comparison of mixing enthalpy value of Brunetti et al.³ and Moelans et al.¹ with the data obtained from this study is presented in Fig. 8. Only

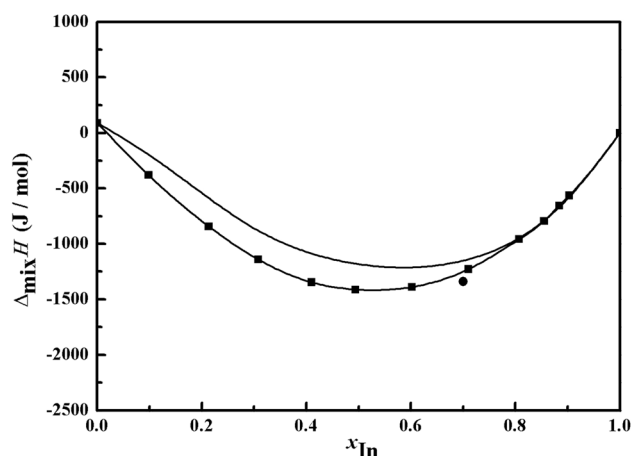


Fig. 8. Comparison of mixing enthalpies of liquid Bi-In-Sn alloys along the cross section: $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$; square, at 767 K (this study), Brunetti et al.³; circle and Moelans et al.¹; bold line.

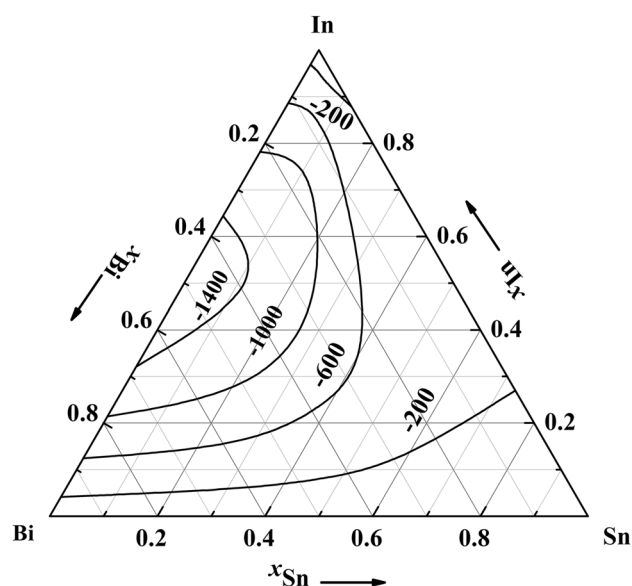


Fig. 9. Iso-enthalpy curves of liquid Bi-In-Sn alloys at 813 K; values are in J/mol.

one of the mixing enthalpy values from Brunetti et al.'s work is compared with this study. It was observed that the mixing enthalpy of Brunetti et al. is slightly more negative as compared to our data. This may be due to a difference in the temperature and composition of the alloy selected for the measurements by Brunetti et al.³ The enthalpies of mixing values calculated based on R-K_Muggianu polynomials using binary and ternary interaction parameters¹ are presented for the comparison in the Fig. 8. It has been observed that calculated enthalpies are very close to the experimental values towards indium-rich compositions and deviates positively for the rest of the compositions of indium.

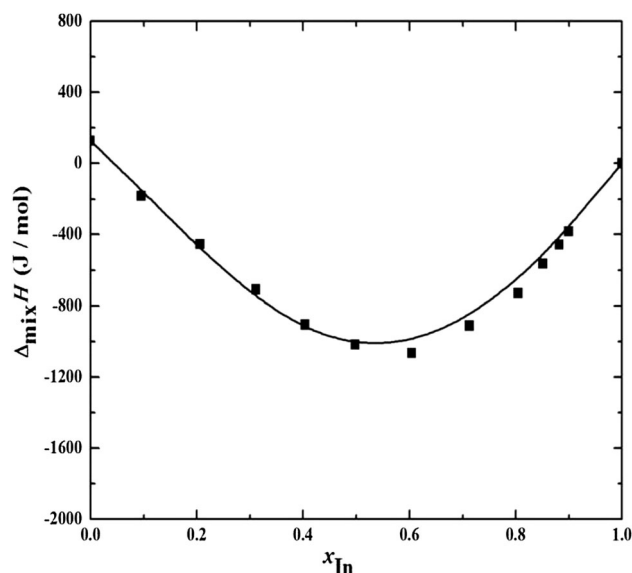


Fig. 10. Comparison of mixing enthalpies between theoretical model; bold line and this study at 813 K for $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ alloys; square.

This may be because the excess entropy in the calculated data is ignored and mixing process has been considered as ideal mixing. Moelans et al.¹ considered only EMF data and no calorimetric data for the optimization of the thermodynamic parameters. These may be why there is a deviation of calculated data from the experimental data.

The iso-enthalpy curves for mixing enthalpy were derived from the molar enthalpy of mixing plot as a function of composition. It is shown in Fig. 9 for four iso-enthalpy values, i.e. -200 J/mol, -600 J/mol, -1000 J/mol and -1400 J/mol. It is observed that the enthalpy of mixing is more negative as the composition of the alloy is close to the Bi-In binary system. This may be due to highest negative enthalpy of mixing values for the Bi-In binary system amongst three binaries.

The iso-enthalpy curves are oriented towards the Bi-In binary system. This indicates that enthalpy of mixing is more negative as the composition of the alloy is close to the Bi-In binary system. This also indicates that stronger interatomic interactions may be present among Bi and In atoms in the ternary sub-lattice.

There are various theoretical models available to interpret the thermodynamic properties of the ternary and higher order systems from the boundary binary data. An important theoretical model has been proposed by Ansara and Dupin.³² It has also been used by Luef et al.³³ using Redlich-Kister-Muggianu polynomial for substitutional solutions. The data from this study were used in the least-squares fit as per Eq. 4.

Table IV. Binary and ternary interaction parameters

Interaction parameter	v	(J/mol)
$L_{\text{Bi-In}}^{(v)}$	0	-6617
	1	133
	2	2066
$L_{\text{Bi-Sn}}^{(v)}$	0	520
	1	235
	2	-157
$L_{\text{In-Sn}}^{(v)}$	3	-119
	0	-1980
	1	-893
$L_{\text{Bi-Sn-In}}^{(v)}$	0	-3874
	1	-12777
	2	24187

$$\Delta_{\text{mix}}H = \sum_i \sum_{j>i} \left[x_i x_j \sum_v L_{ij}^{(v)} (x_i - x_j)^v \right] + x_i x_j x_k \left(L_{ij,k}^0 x_i + L_{ij,k}^1 x_j + L_{ij,k}^2 x_k \right) \quad (4)$$

where $L_{ij}^{(v)}$ ($v = 0, 1, 2, \dots$) are the binary interaction parameters of the boundary binary systems and $L_{ij,k}^{(v)}$ ($v = 0, 1, 2$) are the ternary interaction parameters. The enthalpy of mixing results from the ternary interactions in this system. Ternary enthalpy of mixing for Bi-In-Sn systems for the cross section $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ is represented in Fig. 10 along with least-squares fit curve at 813 K as per Eq. 4.

The ternary interaction parameters are calculated from the list square fit using binary interaction parameters and results from this study. The binary interaction parameters were taken from our unpublished work. The binary and ternary interaction parameters are given in the Table IV. It has been observed that the calculated curve obtained from the binary and ternary interaction parameters are very close to the experimental results for all the studied composition section except some of the values of enthalpy of mixing for Indium rich compositions. The deviations are within the error limit of the experimental results.

SUMMARY AND CONCLUSIONS

Molar enthalpies of mixing for the ternary Bi-In-Sn system were measured using an MHTC 96 Line Evo drop calorimeter by dropping indium into a Bi-Sn alloy bath. The measurements were carried out for three cross sections. The enthalpies of mixing have been observed to be almost temperature independent for the cross sections $(\text{Sn}_{0.33}\text{Bi}_{0.67})_{1-x}\text{In}_x$, $(\text{Sn}_{0.50}\text{Bi}_{0.50})_{1-x}\text{In}_x$ and $(\text{Sn}_{0.67}\text{Bi}_{0.33})_{1-x}\text{In}_x$ in the temperature range of 767–855 K. A comparison of this study with the literature has been attempted by the authors. Iso-enthalpy lines at 813 K shows that all the lines are bent towards the Bi-In binary

system due to more negative enthalpies of mixing values of the Bi-In binary system. The experimental values were fitted by a Redlich–Kister–Muggianu polynomial to find the ternary interaction parameter, and optimized values were obtained.

REFERENCES

1. N. Moelans, K.C. Hari Kumar, and P. Wollants, *J. Alloys Compd.* 360, 98 (2003).
2. V.T. Witusiewicz, U. Hecht, and S. Rex, *J. Alloys Compd.* 428, 115 (2007).
3. B. Brunetti, D. Gozzia, M. Iervolina, V. Piacentea, G. Zanichi, N. Parodi, and G. Borzone, *Calphad* 30, 431 (2006).
4. D.P. Tao, *J. Alloys Compd.* 457, 124 (2008).
5. M.H. Braga, J. Vizdal, A. Kroupa, J. Ferreira, D. Soares, and L.F. Malheiros, *Calphad* 3, 468 (2007).
6. D.V. Malakhov, X.J. Liu, I. Ohnuma, and K. Ishida, *J. Phase Equilib.* 216, 514 (2000).
7. C. Yang, F. Chen, W. Gierlotka, S. Chen, K. Hsieh, and L. Huang, *Mater. Chem. Phys.* 112, 94 (2008).
8. J. Vizdal, M.H. Braga, A. Kroupa, K.W. Richter, D. Soares, L.F. Malheiros, and J. Ferreira, *Calphad* 31, 438 (2007).
9. C. Luef, A. Paul, J. Vizdal, A. Kroupa, A. Kodentsov and H. Ipser, *Monats. Chem.* 137, 381 (2006).
10. Z. Li, S. Knott, and A. Mikula, *Acta Mater.* 55, 2417 (2007).
11. P. Areas, M. Alaoui-Elbelghitib, M. Gambino, and J.P. Brosa, *Thermochim. Acta* 346, 49–56 (2000).
12. C.K. Behera and A. Sonaye, *Thermochim. Acta* 568, 196 (2013).
13. C.K. Behera and M. Shamsuddin, *Thermochim. Acta* 487, 18 (2009).
14. M.R. Kumar, S. Mohan, and C.K. Behera, *J. Electron. Mater.* 45, 4314 (2016).
15. N. Asryan and A. Mikula, *Z. Metallk* 95, 132 (2004).
16. A. Yazawa and K. Koike, *J. Min. Met. Inst. Japan* 184, 1593 (1968).
17. G. Rickert, P. Lamparterand, S. Steeb, and Z. Natur, *Teil A* 31, 711 (1976).
18. W. Oelsen and K.F. Golucke, *Arch. Eisenhüttenwes.* 29, 689 (1958).
19. A. Yazawa, T. Kawashima, and K. Itagaki, *J. Japan Inst. Met.* 32, 1281 (1968).
20. R.L. Sharkey and M.J. Pool, *Metall. Trans. B* 3, 1773 (1972).
21. O.J. Kleppa, *J. Phys. Chem.* 60, 846 (1956).
22. F.E. Wittig and P. Scheidt, *Z. Phys. Chem.* 28, 120 (1961).
23. J.P. Bros and M. Laffitte, *J. Chem. Thermodyn* 2, 151 (1970).
24. R. Hultgren, P.D. Desai, D.T. Hawkins, M. Gleiser, and K.K. Kelley, *ASM International* (Ohio: Metals Park, 1973).
25. C. Luef, H. Flandorfer, and H. Ipser, *Thermochim. Acta* 417, 47 (2004).
26. B.J. Lee, C.S. Ohand, and J.H. Shim, *J. Electron. Mater.* 25, 983 (1996).
27. T.M. Korhonen and J.K. Kivilahti, *J. Electron. Mater.* 27, 149 (1998).
28. N. David, K. Elaissaoui, J.M. Fiorani, J. Hertz, and M. Vilasi, *ThermochimActa* 413, 127 (2004).
29. A. Yakymovych, S. Mudry, C. Luef, and H. Ipser, *Chem. Met. Alloys* 1, 159 (2008).
30. Z. Gregorczyk, L. Stawarzand, and E. Jurzyk, *J. Chem. Thermodyn.* 13, 647 (1981).
31. O. Kubaschewski and C.B. Alcock, *Metallurgical Thermochemistry* 5th edition New York (1979).
32. I. Ansara and N. Dupin, Cost 507 thermo chemical database for light metal alloys, in: European Commission DG X11, European Commission, Luxembourg, (1998).
33. C. Luef, A. Paul, H. Flandorfer, A. Kodentsov, and H. Ipser, *J. Alloys Compd* 391, 67 (2005).

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